

Quiz 1 – Kinematics

Problem 1 (25 pts)

It is one-dimensional motion along the x-axis. The initial position of the object is $x(0) = 15 \text{ m}$. Figure on the right shows the change in the acceleration with time. What is the x-coordinate of the object at time $t = 6 \text{ s}$, if the initial velocity equals to -10 m/s ?

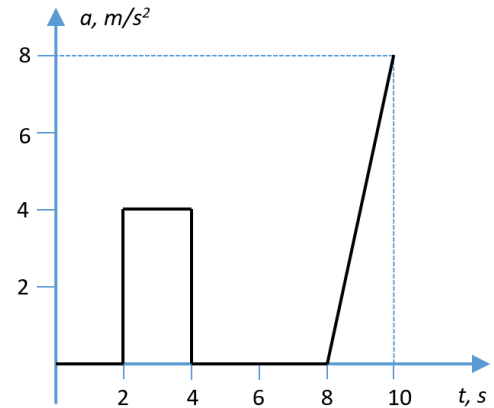


Figure for Problem 1

Problem 2 (25 pts)

The tram starts moving with the constant acceleration a . The passenger waits for 10 s and then catches up the tram by running after with the constant velocity of 12 m/s . What is the maximum acceleration of the tram $a_{\max}$ to make this happen?

Problem 3 (25 pts)

Two balls were thrown from the ground at angles of 30° and 45° to the horizon. The ranges R of the balls were the same. Find the ratio of the initial speeds the balls had at the beginning of their flights. Neglect air resistance.

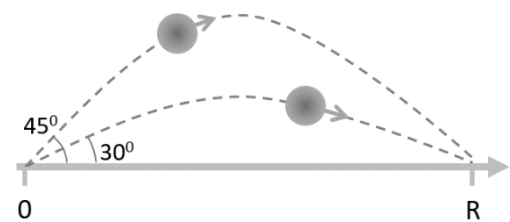


Figure for Problem 3

Problem 4 (25 pts)

A particle moves in uniform circular motion over a horizontal xy plane with the period of 25 s . At one instant, it moves through the point with coordinates $[-1 \text{ m}, 3 \text{ m}]$ with the velocity directed along $\hat{i}$ and the acceleration of $(0 \hat{i} + 6 \hat{j}) \text{ m/s}^2$. Find the x and y coordinates of the center of the circular path.